

17/07/26

Q5

Longest Palindromic Substring

→ Brute force → T.C → $O(n^3)$
→ S.C → $O(n)$

S → "babab"

Output → "bab"
OR
"aba"

① Algorithm:-

- Generate all substring
- Check each substring is palindrome or not
- Store the longest palindrome

② code →

```

public String helper (String str s) {
    StringBuilder ans = new StringBuilder();
    int start = 0;
    for (int s = 0; s < str.length(); s++)
    {
        for (int e = s; e < str.length(); e++)
        {
            bool flag = Palindrome(str.substring(
                (s, e+1)));

```

```

            if (flag && ans.length() < str.substring(
                (s, e+1)).length())
            {

```

```

                ans = new StringBuilder();
                ans.append(str.substring(s, e+1));
            }
        }
    }
    return ans.toString();
}

```

```

public boolean Palindrome (String str) {
    int s = 0; int e = int e = str.length() - 1;
    while (s < e) {
        if (str.charAt(s) != strstr.charAt(e)) {
            return false;
        }
        s++; e--;
    }
    return false, true;
}

```

③ Dry Run →

S = "babad" , ans = ""

①

↓ e
babad
↑ s

flag → Palindrome (b); (T)

i/l (flag == ...) (T) {ans = "" + "b"}

ans → "b"

②

↓ e
babad
↑ s

flag → (T)

i/l (flag == ...) (T)

ans → "ba"

III

s → babad ↖ e

ans ⇒ "ba"

flag → (T)

if (flag == ...) (T)

ans ⇒ "badb"

IV

s → babad ↖ e

ans ⇒ "bab"

flag → (F)

V

s → babad ↖ e

ans ⇒ "bab"

Flag ⇒ (F)

⇒ so this is what we got at 5
Iteration continue ~~to~~ to
get more ans.